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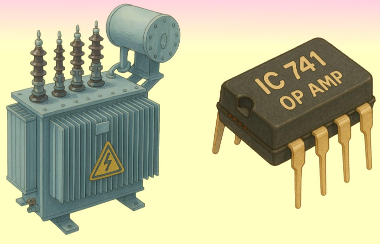
F. E. E. E.

Fundamentals of Electrical & Electronics Engineering Suggestion (2024-2025)

For 2nd Semester of Polytechnic

Top 150 Questions with Complete Answer From All Chapters



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Contents

1. **An ideal current source has-**
(a) zero internal resistance
(b) infinite internal resistance
(c) low value of voltage
(d) large value of current.
Explanation: An ideal current source is designed to supply a constant current regardless of the voltage across its terminals. To achieve this, it has infinite internal resistance, which ensures that the current remains constant even if the load resistance changes.
Answer: (b)

2. **The resistance of a conductor increases if its-**
(a) length increases
(b) area of cross-section increases
(c) length as well as area of cross-section increases
(d) value of specific resistance is kept constant.
Explanation: Resistance of a conductor-
$$R = \rho\frac{L}{A}$$
From this formula, it's clear that resistance (R) increases if the length (L) increases and decreases if the cross-sectional area increases.
Answer: (a)

3. **Rectifier is used to convert-**
(a) DC signal to AC signal
(b) AC signal to Pulsating DC signal
(c) Reducing the amplitude of DC signal
(d) Generation of AC signal
Explanation: A rectifier is an electrical device that converts an alternating current (AC) signal into a pulsating direct current (DC) signal by allowing current to flow in only one direction. This is commonly done using diodes in circuits like half-wave and full-wave rectifiers.
Answer: (b)

4. **For a DC voltage, an inductor-**
(a) is virtually a short circuit.
(b) is an open circuit.
(c) Depends on polarity.
(d) Depends on voltage value.
Explanation: An inductor opposes changes in current due to its property of self-inductance, but for a steady DC voltage, the current remains constant. This means the inductor behaves like a short circuit.
Answer: (a)

5. **Define periodic and non-periodic waveform with one example of each.**
Answer:
Periodic waveform: A waveform that repeats itself at regular intervals of time is called a periodic waveform.
Example: Sine wave, square wave, etc.
Non-periodic waveform: A waveform that does not repeat at regular intervals of time is called a non-periodic (or aperiodic) waveform.
Example: A single pulse or a random noise signal.

6. **Energy is stored by the capacitor in the form of ______.**
Explanation: A capacitor stores energy in the form of an electric field between its plates, known as electrostatic energy.
Answer: electrostatic energy

7. **If two capacitance of 10 μF & 15 μF are connected in parallel then the equivalent capacitance will be ____ μF.**
Explanation: When capacitors are connected in parallel, their total or equivalent capacitance is the sum of the individual capacitances
$$= 10\,\mu F + 15\,\mu F = 25\,\mu F$$
Answer: 25

8. **Give two examples of insulating material.**
Answer: Insulating materials resist the flow of electric current, preventing electrical leakage. Two examples of insulating material: Rubber and Glass.

9. **What is the frequency of a D.C signal?**
Answer: A DC signal has a constant value and does not change over time, so its frequency is zero.

10. **Energy stored by a coil is doubled when its current is increased by percent**
(a) 25 (b) 50.5 (c) 41.4 (d) 100.
Explanation: The energy stored in a coil (inductor) is given by the formula: $U = \frac{1}{2}LI^2$
If the energy is doubled, we can write:
$2U = \frac{1}{2}L(I_{new})^2$
$\Rightarrow 2\left(\frac{1}{2}LI^2\right) = \frac{1}{2}L(I_{new})^2$
$\Rightarrow 2I^2 = I_{new}^2$
$\Rightarrow I_{new} = \sqrt{2}I \approx 1.41I = I141.4\%$
∴ % increased by (141.4-100) = 41.4 %
Answer: (c)

11. **Write the unit of resistivity.**
Answer: The unit of resistivity (ρ) in the International System of Units (SI) is **ohm meter** (Ω·m).

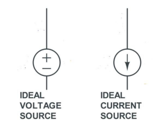
12. **What is the unit of inductance?**
Answer:
Unit of inductance is Henry (H). 1 Henry (H) is the inductance that induces 1 volt when current changes at 1 ampere per second.

13. **What are the types of Rectifier available?**
Answer: Types of Rectifiers are Half-Wave Rectifier and Full-Wave Rectifier. Rectifiers convert AC into DC.

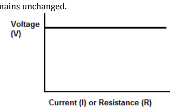
14. **Name the factors affecting the resistance of a conductor.**
Answer: The resistance (R) of a conductor is affected by the following factors:
1) **Length of the Conductor (L):** Resistance is directly proportional to the length of the conductor. R∝L.
2) **Cross-Sectional Area (A):** Resistance is inversely proportional to the cross-sectional area of the conductor. $R \propto \frac{1}{A}$
3) **Resistivity of the Material (ρ):** Resistance depends on the nature of the material, represented by its resistivity. R∝ρ
4) **Temperature:** For most conductors, resistance increases with an increase in temperature due to increased collisions between electrons and atoms. However, for some materials like semiconductors, resistance may decrease with temperature.

15. **What do you mean by capacitance of a capacitor? What is its unit?**
Answer: The capacitance of a capacitor is the ability of the capacitor to store electric charge per unit voltage. It is defined as the ratio of the charge Q stored on the plates of the capacitor to the voltage V applied across the plates: $C = \frac{Q}{V}$
The SI unit of capacitance is the farad (F).
$$1\,F = 1\,C/V$$
Since a farad is a very large unit, practical capacitors are often measured in Microfarad (μF)
$$1\,\mu F = 10^{-6}\,F$$

16. **Define ideal current source and ideal voltage source with proper symbol.**
Answer:
Ideal Current Source: An ideal current source is a two-terminal element that delivers a constant current regardless of the voltage across it or the load connected to it. The output current remains unchanged even if the resistance or impedance of the connected circuit varies.
Ideal Voltage Source: An ideal voltage source is a two-terminal element that maintains a fixed voltage across its terminals, regardless of the current drawn by the connected circuit. The voltage remains constant even if the load resistance or impedance changes.



17. **Draw the load current vs Load voltage curve of an ideal voltage source.**
Answer:
Load Current vs. Load Voltage Curve of an Ideal Voltage Source: An ideal voltage source maintains a constant voltage across its terminals regardless of the current drawn by the load. This means that even if the load current increases, the voltage remains unchanged.



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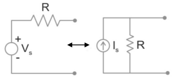
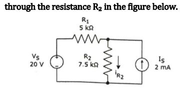
18. **How can you convert a voltage source into current source and vice-versa?**
Answer:
Converting a Voltage Source into a Current Source: A voltage source with a series resistance can be converted into an equivalent current source with a parallel resistance.
Conversion formula: $I_s = \frac{V_s}{R}$
Converting a Current Source into a Voltage Source: A current source with a parallel resistance can be converted into an equivalent voltage source with a series resistance.
Conversion formula: $V_s = I_s R$



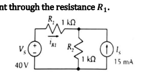
19. **What are active and passive components? Give examples for each.**
Answer:
Active Components: Active components are electronic components that can amplify signals, generate power, or control current flow. They require an external power source to operate and can introduce gain into a circuit.
Examples of Active Components:
1) **Transistors** – used for amplification and switching.
2) **Operational Amplifiers (Op-Amps)** – used in signal processing.
3) **Diodes** (e.g., LED, Zener diode) – allow current to flow in one direction.
Passive Components: Passive components do not require an external power source to operate. They cannot amplify signals but can store, dissipate, or control energy.
Examples of Passive Components:
1) **Resistors** – limit current and control voltage in circuits.
2) **Capacitors** – store and release electrical energy.
3) **Inductors** – store energy in a magnetic field and oppose changes in current.
4) **Transformers** – transfer electrical energy between circuits using electromagnetic induction.

20. **Using source transformation, find out the current through the resistance R_2 in the figure below.**



Solution: Current at voltage source (V_s)
$= \frac{V_s}{R_1} = \frac{20V}{5k\Omega} = 4\,mA$
Since both current sources are in parallel, the equivalent current source is the algebraic sum of the two: $I_{eq} = 4\,mA - 2\,mA = 2\,mA$
The equivalent parallel resistance can be calculated using the formula: $R = \frac{R_1R_2}{R_1+R_2}$
$\Rightarrow R = \frac{5\times7.5}{5+7.5} = \frac{5\times7.5}{12.5} = 3\,k\Omega = 3000\Omega$
∴ The voltage across the parallel combination is:
$V = I_{eq} \times R = 2\,mA \times 3000\Omega = 6\,V$
Finally, the current through R_2
$= I_{R_2} = \frac{V}{R_2} = \frac{6\,V}{7.5\,k\Omega} = 0.8\,mA$ (Ans)

21. **Using Source transformation principle find out the current through the resistance R_1.**



Solution:
Given Circuit Details:
Voltage Source: $V_s = 40$ V
Resistors: $R_1 = 1k\Omega$, $R_2 = 1k\Omega$
Current Source: $I_s = 15$ mA
A voltage source V_s with a series resistance R_1 can be transformed into an equivalent current source:
$$I' = \frac{V_s}{R_1} = \frac{40\,V}{1\,k\Omega} = 40\,mA$$
Since both current sources are in parallel, we sum them up: $I_{eq} = 40\,mA + 15\,mA = 55\,mA$
The total parallel resistance is:
$R_{eq} = \frac{R_1R_2}{R_1+R_2} = \frac{1\times1}{1+1} = \frac{1}{2}\,k\Omega = 500\Omega$
Now, the voltage across the parallel resistors:
$V = I_{eq} \times R_{eq}$
$= 55\,mA \times 500\,\Omega = \frac{55}{1000} \times 500 = 27.5\,V$
∴ Current through R_1:
$I_{R1} = \frac{27.5\,V}{1\,k\Omega} = \frac{27.5}{1000}\,A = 27.5\,mA$ (Ans)

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1. **Which component opposes any change in current?**
a) Resistor b) Capacitor c) Inductor d) Diode
Explanation: An inductor opposes any change in current due to Lenz's Law. It stores energy in the form of a magnetic field and generates an induced electromotive force (EMF) that resists changes in current. This property is called inductive reactance.
Answer: (c)

2. **The property of a material which opposes the creation of magnetic flux is called- a) permeability b) permittivity c) reluctance d) resistance.**
Explanation: Reluctance is the property of a material that opposes the creation of magnetic flux in a magnetic circuit. It is analogous to resistance in an electrical circuit but applies to magnetism. The unit of reluctance is Ampere-turns per Weber (At/Wb).
Answer: (c)

3. **The unit of magnetomotive force is-**
a) ampere-turn b) weber c) mho d) Maxwell.
Explanation: Magnetomotive force (MMF) is the force that drives magnetic flux through a magnetic circuit. It is measured in ampere-turns (At), which is the product of current (in amperes) and the number of turns in a coil.
Answer: (a)

4. **A coil with 500 turns carries a current of 2 A. The value of MMF of the coil is (a) 20 AT (b) 200 AT (c) 1000 AT (d) 55 AT**
Explanation: The magnetomotive force (MMF) is calculated using the formula: $MMF = N \times I$
$N = 500, I = 2A$
$\therefore MMF = 500 \times 2 = 1000\,AT$
Answer: (c)

5. **The energy stored in a magnetic field is given by (L= self inductance, I= current)-**
a) $\frac{1}{2}LI^2$ b) $\frac{1}{2}IL^2$ c) IL^2 d) LI^2
Explanation: The energy (W) stored in the magnetic field of an inductor is given by the formula:
$$W = \frac{1}{2}LI^2$$
Where,
L = Self-inductance (in Henrys, H)
I = Current (in Amperes, A)
Answer: (a)

6. **Eddy current loss is directly proportional to (f = frequency, B_m maximum flux density)- (a) f^2 and B_m^2 (b) f and B_m (c) f and B_m^2 (d) f^2 and B_m.**
Explanation: Eddy current loss (P_e) in a magnetic material is given by the equation: $P_e \propto f^2B_m^2$
Answer: (a)

7. **The direction of induced emf is such that it will oppose the cause to which it is created, is related to ______ law of electromagnetic induction.**
(a) Faraday's (b) Lenz's (c) Ohm's (d) Ampere's
Explanation: Lenz's Law states that- "The direction of the induced EMF (and hence the induced current) is always such that it opposes the change in magnetic flux that caused it."
Answer: (b)

8. **Fleming's Left hand rule gives the direction of ______.**
Explanation: Fleming's Left Hand Rule is used to find the direction of force acting on a current-carrying conductor in a magnetic field.
Answer: force

9. **Conductance in Electric circuit is analogous to ______ in Magnetic Circuit.**
Explanation: In an electric circuit, conductance (inverse of resistance) is analogous to reluctance in a magnetic circuit, which opposes magnetic flux.
Answer: reluctance

10. **A motor converts electrical energy into ______ energy.**
Explanation: An electric motor converts electrical energy into mechanical energy, typically in the form of rotational motion.
Answer: mechanical

11. **What do you mean by permeability?**
Answer:
Permeability (μ): Permeability is the property of a material that determines its ability to allow the formation of a magnetic field within it. It measures how easily a material can support the passage of magnetic flux when exposed to a magnetic field.
Mathematical Expression: $\mu = \frac{B}{H}$
Where:
μ = Permeability (Henry per meter, H/m)
B = Magnetic Flux Density (Tesla, T)
H = Magnetizing Force (A/m)

12. Define: Reluctance

Answer:

Reluctance: Reluctance is the opposition offered by a magnetic circuit to the flow of magnetic flux, similar to electrical resistance in an electric circuit.

$$\text{Reluctance(S)} = \frac{\text{MMF}}{\text{Flux}}$$

13. Define self & mutual inductance.

Answer:

Self-Inductance: Self-inductance is the property of a coil (or circuit) by which it opposes any change in its own current by inducing a voltage (emf) in itself. It is denoted by L, and the unit is henry (H).

Example: An inductor in a DC circuit resisting sudden changes in current.

Mutual Inductance: Mutual inductance is the property where a change in current in one coil induces an emf in a nearby coil due to magnetic coupling.

It is denoted by M, and the unit is also henry (H).

Example: Transformer, where the primary coil induces voltage in the secondary coil.

14. Write down the similarities between magnetic & electric circuit.

Answer:

Similarities between magnetic & electric circuit:

1) **Both circuits have a driving force**: Voltage in electric circuits and Magnetomotive Force (MMF) in magnetic circuits.
2) **Both involve a flow**: Electric current in an electric circuit and Magnetic flux in a magnetic circuit.
3) **Both face opposition to flow**: Resistance in electric circuits and Reluctance in magnetic circuits.
4) **Both follow similar mathematical laws**: Ohm's Law in electric circuits and Magnetic Ohm's Law in magnetic circuits.
5) **Both require a closed path**: Electric circuits need a closed conducting path, while magnetic circuits require a closed loop (magnetic core).

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15. Define Magnetic Flux Density and MMF.

Answer:

Magnetic Flux Density (B): Magnetic Flux Density is defined as the amount of magnetic flux passing through a unit area perpendicular to the direction of the magnetic field. It represents the strength and direction of the magnetic field in a given region.

Formula: $B = \frac{\phi}{A}$, SI unit: Weber/m² or Tesla (T)

Magnetomotive Force (MMF): Magnetomotive Force (MMF) is the force that drives magnetic flux through a magnetic circuit. It is analogous to the electromotive force (EMF) in electrical circuits.

Formula: $F_m = NI$

N = Number of turns in the coil

I = Current flowing through the coil (A)

SI Unit: Ampere-Turns (A·t)

16. **State Faraday's Laws of Electromagnetic Induction.**

Answer:

Faraday's laws of electromagnetic induction:

First law: It states that whenever there is a change in magnetic flux associated with a coil, EMF is induced in that coil.

Second law: It states that the magnitude of EMF induced in the coil is directly proportional to the rate of change of magnetic flux associated with that coil.

Mathematically, it can be expressed as $\varepsilon = -\frac{d\phi}{dt}$, where ε is the induced EMF and $\frac{d\phi}{dt}$ is the rate of change of magnetic flux. For n number of turns in the coil, the expression is given as $\varepsilon = -n\frac{d\phi}{dt}$.

17. Define Lenz's Law?

Answer: Lenz's Law states that the direction of an induced electromotive force (emf) or current in a conductor is always such that it opposes the change in magnetic flux that caused it.

This law is a consequence of the law of conservation of energy and is fundamental in electromagnetic induction.

Mathematically, it is expressed as:

$$e = -N\frac{d\phi}{dt}$$

e is the induced emf,

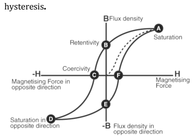
ϕ is the magnetic flux,

N is number of turns in coil,

The negative sign indicates opposition to the change in flux.

18. Draw and explain hysteresis loop for magnetic material.

Answer: A hysteresis loop represents the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a magnetic material. It demonstrates how the material retains magnetism even after the external magnetic field is removed. This phenomenon is called magnetic hysteresis.



Hysteresis Loop Explanation:

- **Point A (Saturation)**: The material reaches its maximum magnetization when all magnetic domains align.
- **Point B (Retentivity)**: After removing the external field, the material retains some magnetism, called **residual magnetism** or **retentivity**.
- **Point C (Coercivity)**: A reverse magnetic field is applied to reduce the magnetic flux to zero. The required field strength is called **coercive force**.
- **Point D (Negative Saturation)**: Applying a stronger reverse field aligns the domains in the opposite direction, causing negative saturation.
- **Energy Loss**: The area within the loop represents **hysteresis loss** due to the energy used in reversing magnetic domains.
- **Narrow Loop** → Low energy loss (e.g., soft iron for transformers)
- **Wide Loop** → High energy retention (e.g., steel for permanent magnets)

19. Define: Hysteresis loss

Answer:

Hysteresis Loss: Hysteresis loss is the energy loss in a magnetic material due to the repeated magnetization and demagnetization cycles when subjected to an alternating magnetic field. It is caused by domain realignment within the material and appears as heat.

20. Discuss the causes of hysteresis loss.

Answer:

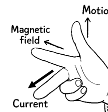
Causes of Hysteresis Loss:

1) **Domain Realignment**: Energy is used to rotate and reorient magnetic domains during magnetization and demagnetization.
2) **Internal Friction**: Domain movement faces resistance, causing energy loss as heat.
3) **Defects and Impurities**: Structural imperfections hinder domain motion, increasing energy loss.
4) **Domain Wall Pinning**: Domain walls get stuck at impurities, requiring extra energy to move.
5) **Material Properties**: Hard magnetic materials have higher hysteresis loss, while soft materials have lower loss.

21. State Fleming's Right hand Rule?

Answer:

Fleming's Right-Hand Rule: Stretch the thumb, index finger, and middle finger of your right hand mutually perpendicular to each other:



Thumb → Direction of motion (force) of the conductor

Index finger → Direction of the magnetic field (North to South)

Middle finger → Direction of the induced current

This rule helps predict the direction of current produced in generators and other induction-based devices.

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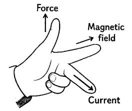
22. State Fleming's left hand rule and its application.

Answer:

Fleming's Left-Hand Rule: Fleming's Left-Hand Rule is used to determine the direction of force acting on a current-carrying conductor placed in a magnetic field. It states that: If you stretch the thumb, forefinger, and middle finger of your left hand mutually perpendicular to each other:



- **Forefinger** : Points in the direction of the magnetic field (B)
- **Middle finger** : Points in the direction of the current (I)
- **Thumb** : Points in the direction of the force (F) or motion of the conductor

Application of Fleming's Left-Hand Rule:

1) **Electric Motors**: It helps in determining the direction of motion of the rotor in motors like DC motors, where current-carrying conductors experience a force due to a magnetic field.
2) **Magnetic Levitation**: Used to find the force direction acting on current-carrying coils, which helps in levitation.
3) **Loudspeakers**: The movement of the diaphragm in loudspeakers is determined using this rule, where an alternating current interacts with a magnetic field.

23. **Derive the expression for the energy stored in the magnetic field of an inductor.**

Answer: An inductor stores energy in the form of a magnetic field when current flows through it.

Step 1: Induced EMF in an Inductor

From Faraday's Law, the voltage across an inductor is:

$$V = L\frac{dL}{dt}$$

L = Inductance (H)

I = Current (A)

Step 2: Power Supplied to the Inductor

Power (P) is given by: $P = VI$

$\Rightarrow P = LI\frac{dL}{dt}$ (Substituting V)

Step 3: Energy Stored in the Inductor

Energy supplied in an infinitesimal time interval (dt):

$dW = Pdt = LI\frac{dL}{dt}dt = LIdI$

$\Rightarrow W = \int_0^I LI\,dI = L\left[\frac{I^2}{2}\right]_0^I = \frac{1}{2}LI^2$

Final expression: $W = \frac{1}{2}LI^2$

24. A rectangular shape iron core has an air gap of 0.01cm. The mean length of the flux path through iron is 39.99 cm. The relative permeability of iron is 2000. The coil has 1000 turns. The cross-sectional area of the core is 9 cm². Calculate the current required to produce a flux of 1 mWb in the core.

Solution: Given,

Air gap $L_a = 0.01\,cm = 10^{-4}\,m$

Relative permeability $\mu_r = 2000$

Number of turns of the coil $T = 1000$

Cross-sectional area of the core

$$A = 9\,cm^2 = 9 \times 10^{-4}\,m^2$$

Flux $\phi = 1\,mWb = 10^{-3}\,Wb$

Length of the flux path through iron

$$L_m = 39.99\,cm = 39.99 \times 10^{-2}\,m$$

Permeability of air $\mu_0 = 4\pi \times 10^{-7}$

Total reluctance

= Reluctance of air gap + Reluctance of iron core

$= \frac{L_a}{\mu_0 A} + \frac{L_m}{\mu_r \mu_0 A}$

$= \frac{1}{\mu_0 A}\left(L_a + \frac{L_m}{\mu_r}\right)$

$= \frac{1}{4\pi \times 10^{-7} \times 9 \times 10^{-4}}\left(10^{-4} + \frac{39.99 \times 10^{-2}}{2000}\right)$

$= 2.65 \times 10^5\,AT/Wb$

$MMF = I \times T = I \times 1000$

$\text{Flux} = \frac{\text{MMF}}{\text{Raluctance}}$

$\Rightarrow 10^{-3} = \frac{I \times 1000}{2.65 \times 10^5}$

$\Rightarrow I = \frac{10^{-3} \times 2.65 \times 10^5}{1000} = 0.265\,A$ (Ans)

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1. **The duration of one cycle known as- a) waveform b) peak value c) instantaneous value d) period.**

Explanation: The period is the duration of one complete cycle of a waveform. It is usually denoted by T and measured in seconds (s). The period is the reciprocal of frequency (T = 1/f), where frequency is measured in Hertz (Hz).

Answer: (d)

2. **The unit of inductive reactance is- (a) Henry (b) Farad (c) Ohm (d) Joule**

Explanation: Inductive reactance ($X_L = 2\pi fL$) is the opposition offered by an inductor to AC current. Since reactance is a type of resistance in AC circuits, it is measured in Ohms (Ω), just like electrical resistance.

Answer: (c)

3. **Inductive reactance of a coil of inductance 0.2 H at 50 Hz is (a) 62.8 Ω (b) 628 Ω (c) 0.2 Ω (d) 10 Ω.**

Explanation: Given, inductance $L = 0.2\,H$

Frequency $f = 50\,Hz$

∴ The inductive reactance $X_L = 2\pi fL$

$\Rightarrow X_L = 2 \times 3.1416 \times 50 \times 0.2 = 62.83\,\Omega$

Answer: (a)

4. **The power factor of a purely resistive circuit is (a) zero (b) unity (c) lagging (d) leading.**

Explanation: In a purely resistive circuit, the voltage and current are in phase with each other, meaning there is no phase difference ($\phi = 0$) between them.

∴ Power factor $= \cos\phi = \cos 0 = 1$

Answer: (b)

5. **The maximum value of sine wave current is 100 A, the rms value of current in the circuit will be- a) 70.7 A b) 100 A c) 50 A d) 70.7 mA.**

Explanation: For a sine wave, the RMS (Root Mean Square) value is related to the maximum (peak) value by: $I_{rms} = \frac{I_{peak}}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.71\,A$

Answer: (a)

6. **Peak factor for a sine wave is- a) 1 b) 1.414 c) 1.11 d) 2**

Explanation: The peak factor (also called crest factor) is the ratio of the peak value to the RMS (root mean square) value of a waveform.

$\text{Peak factor} = \frac{V_{peak}}{V_{rms}} = \frac{V_{peak}}{\frac{V_{peak}}{\sqrt{2}}} = \sqrt{2} = 1.414$

Answer: (b)

7. **A sinusoidal voltage is represented by $V = 141.4\sin\left(\frac{314.18\pi}{2}\right)$. Its rms value of voltage is- (a) 141.4 V (b) 100 V (c) 87.92 V (d) 200 V.**

Explanation: The peak value of the sinusoidal voltage is given as: $V_0 = 141.4\,V$

The RMS (Root Mean Square) value-

$$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{141.4}{1.414} \approx 100\,V$$

Answer: (b)

8. **Two sinusoidal emfs are given as- $e_1 = A\sin\left(\omega t + \frac{\pi}{4}\right)$ and $e_2 = B\sin\left(\omega t - \frac{\pi}{6}\right)$ The phase difference between the two quantities in degrees is- a) 15 b) 45 c) 75 d) 30.**

Explanation: The phase difference between the two sinusoidal EMFs is calculated as:

$\phi = \frac{\pi}{4} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{4} + \frac{\pi}{6} = 45° + 30° = 75°$

Answer: (c)

9. **When sinusoidal voltage is applied across a pure capacitor the current in the capacitor-**
a) lags applied voltage by 90°
b) leads applied voltage by 90°
c) is in phase with applied voltage
d) is in phase opposition with applied voltage

Explanation: When a sinusoidal voltage is applied across a pure capacitor, the current leads the voltage by 90 degrees. This means the current reaches its peak value a quarter cycle before the voltage does.

Answer: (b)

10. **The current through a series RLC circuit under resonance condition will be (a) V/R (b) V/Xc (c) V/X_L (d) none of these**

Explanation: In a series RLC circuit, resonance occurs when $X_L = X_c$

Impedance $Z = R + j(X_L - X_c) = R + 0 = R$

Using Ohm's Law: $I = \frac{V}{Z} = \frac{V}{R}$

Answer: (a)

11. **The time constant of a series R-C circuit is given by ________.**

Explanation: The time constant (τ) of a resistor-capacitor circuit is the product of resistance (R) and capacitance (C), determining how quickly the circuit charges or discharges.

Answer: <u>τ = R × C</u>

12. Define the term form factor.

Answer: Form factor is the ratio of the RMS value to the average value of an AC waveform. For a sinusoidal wave, form factor = 1.11.

13. Calculate the time period of a periodic signal whose frequency is 60Hz.

Answer: The time period (T) is the reciprocal of frequency (f). So $T = \frac{1}{f} = \frac{1}{60} = 0.0167$ s (Ans)

14. In an ac circuit '$VI\sin\theta$' represents- a) Active power b) Reactive power c) Apparent power d) True power.

Explanation: In an AC circuit, different types of power are defined as:

1) Apparent Power, $S = VI$
   Measured in Volt-Amperes (VA)
2) Active (Real) Power $P = VI\cos\theta$
   Measured in Watts (W)
3) Reactive Power, $Q = VI\sin\theta$
   Measured in Volt-Amperes Reactive (VAR)

Answer: (b)

15. Weber is the unit of ______.

Explanation: The unit of magnetic flux is Weber (Wb), which is equal to one volt-second.

Answer: <u>magnetic flux</u>

16. What is the unit of reactive Power?

Answer: The unit of reactive power is VAR (Volt-Ampere Reactive). Reactive power (Q) is the power stored and released in inductors and capacitors.

17. Define: RMS value

Answer:

RMS Value (Root Mean Square Value): The RMS value of an alternating current (AC) or voltage is the equivalent DC value that produces the same heating effect in a resistive circuit. It is given by:

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}, \quad I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

18. Define: Frequency

Answer:

Frequency (f): Frequency is the number of complete cycles of an alternating quantity (such as voltage or current) per second. It is measured in Hertz (Hz).

$$f = \frac{1}{T}$$

where T is the time period of one complete cycle.

19. Define: Flux density

Answer:

Flux Density (B): Flux density is the amount of magnetic flux (ϕ) per unit area of a surface perpendicular to the magnetic field. It is measured in Tesla (T).

$$B = \frac{\phi}{A}$$

ϕ = Magnetic flux (Weber, Wb)

A = Cross-sectional area (m²)

20. Define: Form factor

Answer:

Form Factor: The form factor is the ratio of the RMS value to the average value of an alternating waveform.

$$\text{Form factor} = \frac{\text{RMS value}}{\text{Average value}}$$

For a sinusoidal waveform,

$$\text{Form factor} = \frac{0.707\,V_m}{0.637\,V_m} = 1.11$$

21. Calculate the RMS and average values of a purely sinusoidal current having peak value 15A.

Solution: Given, peak current $I_{peak} = 15\,A$

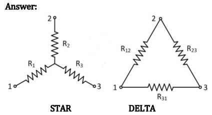
∴ RMS value, $I_{rms} = \frac{I_{peak}}{I_{rms}} = \frac{15}{\sqrt{2}} = 10.61\,A$ (Ans)

The average value of a pure sine wave over a full cycle is zero, but over half cycle: $I_{avg} = \frac{2I_{peak}}{\pi}$

$$\Rightarrow I_{avg} = \frac{2 \times 15}{\pi} \approx 9.55\,A \text{ (Ans)}$$

22. Draw STAR & DELTA connections. Also write voltage and current relations for both types.

Answer:



Voltage & Current Relations:

Parameter	Star (Y) Connection	Delta (Δ) Connection
Line Voltage (V_L)	$V_L = \sqrt{3}V_P$	$V_L = V_P$
Line Current (I_L)	$I_L = I_P$	$I_L = \sqrt{3}I_P$

I_P = Phase current, V_P = Phase voltage

23. **For sinusoidally varying alternating quantity, state the relation between**
(i) rms value and maximum value,
(ii) average value and maximum value.

Answer: For a sinusoidally varying alternating quantity

1) **RMS Value and Maximum Value**

$$V_{rms} = \frac{V_{max}}{\sqrt{2}} \quad \text{and} \quad I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

RMS (Root Mean Square) value is 0.707 times the peak (maximum) value.

2) Average Value and Maximum Value (For a full half-cycle)

$$V_{avg} = \frac{2V_{max}}{\pi} \quad \text{and} \quad I_{avg} = \frac{2I_{max}}{\pi}$$

The average value over a half-cycle is 0.637 times the peak value.

24. Define active power and reactive power.

Answer:

Active Power (Real Power) P: Active power is the actual power consumed by a circuit to perform useful work, such as producing heat, motion, or light. It is measured in watts (W) and is given by:

$$P = V_{rms}I_{rms}\cos\phi$$

Reactive Power Q: Reactive power is the power that oscillates between the source and reactive elements (inductors and capacitors) without being converted into useful work. It is measured in volt-amperes reactive (VAR) and is given by:

$$Q = V_{rms}I_{rms}\sin\phi$$

25. What is power factor? Write the value of power factor for purely resistive circuit.

Answer:

Power Factor: The power factor (PF) is the cosine of the phase angle between the voltage and the current in an AC circuit. It indicates how effectively the electrical power is being converted into useful work. It is given by:

$$\text{Power Factor} = \frac{\text{Real Power (P)}}{\text{Apparent Power (S)}} = \frac{VI\cos\phi}{VI} = \phi$$

where ϕ is the phase angle between voltage and current.

In a purely resistive circuit, voltage and current are in phase ($\phi = 0°$), so power factor $= \cos 0° = 1$

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26. **Show that in a pure capacitive circuit the current leads the applied voltage by 90 degree. Also show the phasor diagram for the same.**

Answer: An ideal capacitor with capacitance C is connected to an alternating voltage source:

$$V = V_m\sin(\omega t)$$

V_m is the peak voltage, t is time.

$\omega = 2\pi f$ is the angular frequency,

From capacitor theory, the current III in a capacitor is related to the voltage by: $I = C\frac{dV}{dt}$

$\Rightarrow I = C\frac{d}{dt}\{V_m\sin(\omega t)\}$

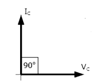
$\Rightarrow I = CV_m\omega\cos(\omega t)$

$\Rightarrow I = I_m\sin\left(\omega t + \frac{\pi}{2}\right)$ [taking $CV_m\omega = I_m$]

Phase difference between current and voltage:

$\left(\omega t + \frac{\pi}{2}\right) - (\omega t) = \frac{\pi}{2}$

Thus, in a purely capacitive AC circuit, the current leads the voltage by π/2 radians (or 90° in phase).



phasor diagram

27. Show that current in a pure inductive circuit lags the voltage by 90 degree.

Answer: A pure inductive circuit consists of an inductor L connected to an AC voltage source:

$$v = V_m\sin(\omega t)$$

where:

V_m is the peak voltage,

ω is the angular frequency,

t is time.

According to Faraday's Law, the voltage across an inductor is given by: $v = L\frac{di}{dt}$,

$\Rightarrow di = \frac{v}{L}dt$

$\Rightarrow di = \frac{V_m\sin(\omega t)}{L}dt$

$\Rightarrow i = \int\frac{V_m\sin(\omega t)}{L}dt$

$\Rightarrow i = -\frac{V_m}{\omega L}\cos(\omega t)$

$\Rightarrow i = -\frac{V_m}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$

The current i is sinusoidal, just like the voltage.

However, it has a phase shift of $-\frac{\pi}{2}$ (i. e. 90° lag)

28. A R-L-C series circuit comprises of resistance of 15Ω inductance 0.15 H and capacitance of 100 µF is connected across a 230V, 50 Hz supply. Calculate current, power factor and power consumed by the circuit.

Solution:

Given, resistance $R = 15\ \Omega$

Inductance $L = 0.15\ H$

Capacitance $C = 100\ \mu F = 100 \times 10^{-6}\ F$

Supply voltage $V = 230\ V$

Frequency $f = 50\ Hz$

Now, inductive reactance $X_L = 2\pi f L$

$\Rightarrow X_L = 2 \times 3.1416 \times 50 \times 0.15 = 47.12\ \Omega$

Capacitive reactance $X_C = \frac{1}{2\pi f C}$

$$X_C = \frac{1}{2 \times 3.1416 \times 50 \times 100 \times 10^{-6}} = 31.83\ \Omega$$

$\therefore$ Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\Rightarrow Z = \sqrt{15^2 + (47.12 - 31.83)^2} = 21.45\ \Omega$

Current $I = \frac{V}{Z} = \frac{230}{21.45} = 10.73\ A$ (Ans)

Power factor $\cos\phi = \frac{R}{Z} = \frac{15}{21.45} = 0.7$ (Ans)

Power consumed,

$$P = VI\cos\phi$$

$\Rightarrow P = 230 \times 10.73 \times 0.7 = 1727.53\ W$ (Ans)

29. A coil of resistance 20 Ω and inductance 100 mH is connected in series with a capacitance of 40 µF across 100 V, 50 Hz ac supply. Calculate
(i) the magnitude of current,
(ii) power factor,
(iii) phase angle between voltage and current,
(iv) voltage across each element.

Answer:

Given, resistance $R = 20\ \Omega$,

Inductance $L = 100\ mH = 0.1\ H$

Capacitance $C = 40\ \mu F = 40 \times 10^{-6}\ F$

Supply voltage $V = 100\ V$

Frequency $f = 50\ Hz$

$\therefore$ Inductive reactance $X_L = 2\pi f L$

$= 2 \times 3.142 \times 50 \times 0.1 = 31.42\ \Omega$

Capacitive reactance $X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2 \times 3.142 \times 50 \times 40 \times 10^{-6}} = 79.57\ \Omega$

$\therefore$ Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\Rightarrow Z = \sqrt{20^2 + (31.42 - 79.57)^2} = 52.14\ \Omega$

(i) Magnitude of Current $I = \frac{V}{Z}$

$$= \frac{100}{52.14} = 1.92\ A \text{ (Ans)}$$

(ii) Power factor $\cos\phi = \frac{R}{Z} = \frac{20}{52.14} \approx 0.384$ (Ans)

(iii) Phase angle $\phi = \cos^{-1}(0.384) = 67.42°$ (Ans)

(iv) Voltage across each element:

Voltage across resistor $V_R = IR$

$= 1.92 \times 20 = 38.4\ V$ (Ans)

Voltage across inductor $V_L = IX_L$

$= 1.92 \times 31.42 = 60.3\ V$ (Ans)

Voltage across capacitor $V_C = IX_C$

$= 1.92 \times 79.57 = 152.7\ V$ (Ans)

30. A series RLC circuit having resistance of 10 Ω, inductance of 0.1 H and capacitance of 150 µF are connected in series across 230 V, 50 Hz AC supply. Calculate:
i) Impedance
ii) current,
iii) power factor,
iv) Active power,
v) Voltage drop across inductor.

Answer:

Given, resistance $R = 10\ \Omega$,

Inductance $L = 0.1\ H$

Capacitance $C = 150\ \mu F = 150 \times 10^{-6}\ F$

Supply voltage $V = 230\ V$

Frequency $f = 50\ Hz$

$\therefore$ Inductive reactance $X_L = 2\pi f L$

$= 2 \times 3.142 \times 50 \times 0.1 = 31.42\ \Omega$

Capacitive reactance $X_C = \frac{1}{2\pi f C}$

$= \frac{1}{2 \times 3.142 \times 50 \times 150 \times 10^{-6}} = 21.22\ \Omega$

i) Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\Rightarrow Z = \sqrt{10^2 + (31.42 - 21.22)^2} = 14.28\ \Omega$

ii) Magnitude of Current $I = \frac{V}{Z}$

$$= \frac{230}{14.28} = 16.11\ A \text{ (Ans)}$$

iii) Power factor $\cos\phi = \frac{R}{Z} = \frac{10}{14.28} = 0.7$ (Ans)

iv) Active power $= VI\cos\phi$

$= 230 \times 16.11 \times 0.7 = 2593.71\ W$ (Ans)

v) Voltage across inductor $V_L = IX_L$

$= 16.11 \times 31.42 = 506.18\ V$ (Ans)

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1. **Voltage per turn in both primary and secondary windings of a transformer is-**
   (a) high in high voltage winding
   (b) low in low voltage winding
   (c) same
   (d) can't be determined.
   Explanation: In an ideal transformer, the voltage per turn is the same for both the primary and secondary windings. This relationship is based on the principle of electromagnetic induction, which states that the induced voltage per turn in both windings must be equal to maintain the magnetic flux linkage.
   Answer: (c)

2. **Transformer rating are given in**
   (a) kVA (b) HP (c) KVAR (d) kW
   Explanation: Transformers are rated in kVA rather than kW because they do not account for the power factor of the load. The apparent power (in kVA) represents the total power supplied by the transformer, including both active power (kW) and reactive power (kVAR). Since the transformer handles both types of power, its rating is given in kVA.
   Answer: (a)

3. **Which of the following does not change in a transformer? (a) Current (b) Voltage (c) Frequency (d) All of the above.**
   Explanation: A transformer works on the principle of electromagnetic induction, where it changes the voltage and current while keeping the power (neglecting losses) and frequency constant.
   Answer: (c)

4. **A transformer transforms –**
   a) Power b) Voltage c) current d) frequency
   Explanation: A transformer primarily transforms voltage and current based on the turns ratio between its primary and secondary windings.
   Answer: (b), (c)

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5. **The flux in transformer core-**
   (a) increases with load,
   (b) decreases with load,
   (c) remains constant irrespective of load,
   (d) none of the above.
   Explanation: The load current only affects the secondary and primary winding currents, but the flux in the core remains constant as long as the supply voltage and frequency do not change.
   Answer: (c)

6. **In an auto-transformer, power is transferred, through**
   (a) Conduction process only
   (b) Induction process only
   (c) Both conduction and induction processes
   (d) Mutual coupling.
   Explanation: In an auto-transformer, power is transferred both by conduction (through the common winding) and induction (via mutual induction between primary and secondary). This makes it more efficient compared to a conventional transformer, which transfers power only by induction.
   Answer: (c)

7. **Direction of rotation of the motor armature/rotor can be determined by applying Fleming's left hand rule. (True/False)**
   Explanation: Fleming's Left Hand Rule helps determine the direction of force (and thus rotation) in a motor by aligning the thumb (force), forefinger (magnetic field), and middle finger (current) at right angles to each other.
   Answer: True

8. **A single phase 50Hz transformer has 80 turns on the primary winding & 400 turns on the secondary winding. If the primary winding is connected to a 240 volt, 50 Hz supply, determine the e.m.f induced in the secondary winding.**
   Solution: Given, $N_p = 80, N_s = 400, V_p = 240\ V$
   We can solve this using the transformer turns ratio formula, $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (as usual notation)
   $$\Rightarrow V_s = \frac{V_p N_s}{N_p} = \frac{240 \times 400}{80} = 1200\ V$$
   Induced e.m.f. in the secondary winding = 1200 V

9. **A 10 KVA, 1100/400 V, 50 Hz single phase transformer has 100 turns in the secondary winding. The number of turns in primary winding is ________. (Fill in the blank)**
   Solution: We can solve this using the transformer turns ratio formula, $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (as usual notation)
   $$\Rightarrow N_p = \frac{V_p N_s}{V_s} = \frac{1100 \times 100}{400} = 275$$
   Answer: 275

10. **Transformer cores are laminated in order to ____.**
    Explanation: Laminating the transformer core reduces the circulating eddy currents, thereby minimizing the power loss and improving efficiency.
    Answer: reduce eddy current losses

11. **What type of DC Motor used in traction?**
    Answer: DC series motor is used in traction. It provides high starting torque, making it ideal for railway locomotives and electric vehicles.

12. **Name different types of DC motors.**
    Answer:

Type of DC Motor	Applications
Separately Excited	Research labs, precision control
Shunt DC Motor	Fans, pumps, conveyors
Series DC Motor	Trains, cranes, electric vehicles
Compound DC Motor	Elevators, presses, rolling mills
Permanent Magnet DC Motor (PMDC)	Robotics, car wipers, toys

13. **Define a Transformer.**
    Answer: A transformer is a static electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. It is used to increase (step-up) or decrease (step-down) AC voltage levels while maintaining the same frequency.
    Key Points:
    1) Works only with AC (alternating current).
    2) Based on the principle of mutual induction.
    3) Consists of primary and secondary windings wound on a common magnetic core.

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14. **On which principle the transformer works?**
    Answer: A transformer works on the principle of electromagnetic induction. It transfers electrical energy between circuits through mutual induction.

15. **What do you mean by voltage transformation ratio of a transformer?**
    Answer: The voltage transformation ratio (also called the turns ratio) of a transformer is the ratio of the voltage in the secondary winding to the voltage in the primary winding. It is given by the formula:
    $$K = \frac{V_s}{V_p} = \frac{N_s}{N_p}$$
    where:
    K = Voltage transformation ratio
    V_s = Secondary voltage
    V_p = Primary voltage
    N_s = Number of turns in the secondary coil
    N_p = Number of turns in the primary coil
    For step-up Transformer $K > 1$
    For step-down Transformer $K < 1$

16. **What is mutual inductance?**
    Answer: Mutual inductance (M) is the property of two coils or circuits in which a change in current in one coil induces an electromotive force (emf) in the other. This occurs due to the magnetic field generated by one coil linking with the other.
    Mathematically, mutual inductance is given by:
    $$\mathrm{M} = \frac{\text{Induced emf in secondary coil(e)}}{\text{Rate of change of current in primary coil}\left(\frac{\mathrm{di}}{\mathrm{dt}}\right)}$$
    Mutual inductance is the working principle behind transformers, wireless charging, and induction coils.

17. **Write four main parts of DC motor.**
    Answer: The four main parts of a DC motor are:
    1) **Stator**: The stationary part of the motor that provides a constant magnetic field using either permanent magnets or field windings.
    2) **Rotor (Armature)**: The rotating part of the motor, consisting of coils of wire that carry current and interact with the magnetic field to produce motion.
    3) **Commutator**: A split-ring device attached to the rotor that reverses the direction of current flow in the armature windings, ensuring continuous rotation.
    4) **Brushes**: Conductive carbon or graphite blocks that maintain electrical contact between the external circuit and the rotating commutator.

18. **Write down the working principle of two winding transformer.**
    Answer: The working principle of a two-winding transformer is based on Faraday's law of electromagnetic induction. It operates on the principle of mutual induction between two windings (primary and secondary) to transfer electrical energy from one circuit to another.
    Working Principle:
    1) **AC Supply to Primary Winding:** When an alternating voltage is applied to the primary winding, an alternating current flows through it, producing a time-varying magnetic flux in the transformer core.
    2) **Magnetic Flux Linkage:** The changing magnetic flux created by the primary winding is linked to the secondary winding through the magnetic core.
    3) **Electromagnetic Induction in Secondary Winding:** According to Faraday's Law, this changing magnetic flux induces an electromotive force (EMF) in the secondary winding.
    4) **Energy Transfer:** If the secondary winding is connected to a load, a current flows through it, transferring electrical energy from the primary to the secondary circuit.
    5) **Voltage Transformation:** The ratio of the primary to secondary voltage is determined by the turns ratio of the transformer:
    $$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$
    where:
    V_p and V_s are the primary and secondary voltages,
    N_p and N_s are the number of turns in the primary and secondary windings.
    This allows the transformer to step up (increase) or step down (decrease) the voltage as required.

19. **Give the applications of DC series motor.**
    Answer:
    Applications of DC Series Motor:
    1) **Electric Locomotives:** Used in trains and trams where high torque is needed for starting and acceleration.
    2) **Cranes and Hoists**: Ideal for lifting heavy loads due to their strong torque characteristics.
    3) **Electric Vehicles (EVs)**: Used in older electric cars, forklifts, and trolleys for efficient torque handling.
    4) **Elevators and Lifts**: Provides smooth and powerful operation in vertical transportation systems.
    5) **Conveyor Systems**: Used in industries for moving heavy materials efficiently.
    6) **Rolling Mills and Presses**: Employed in metal industries where variable speed and high torque are required.

20. **Derive the emf equation of a transformer.**
    Answer:
    Considerations:
    1) A transformer consists of a primary coil and a secondary coil wound around a common core.
    2) An alternating voltage applied to the primary winding produces a time-varying magnetic flux (ϕ) in the core.
    3) This varying flux links both windings and induces an EMF in each according to Faraday's Law.

    The EMF induced in a coil is given by (with usual notation):
    $$E = -N\frac{d\phi}{dt}$$
    Assume that the flux ϕ varies sinusoidally:
    $$\phi = \phi_m \sin(\omega t)$$
    Taking the derivative of ϕ:
    $$\frac{d\phi}{dt} = \omega\phi_m \cos(\omega t)$$
    The RMS value of induced EMF is found using the peak value: $E_{peak} = N\omega\phi_m$
    Since for a sinusoidal waveform, RMS value is given by:
    $$E_{rms} = \frac{E_{peak}}{\sqrt{2}}$$
    $$\Rightarrow E = \frac{N\omega\phi_m}{\sqrt{2}}$$
    Substituting $\omega = 2\pi f$
    $$\Rightarrow E = \frac{N(2\pi f)\phi_m}{\sqrt{2}}$$
    $$\Rightarrow E = 4.44 f N\phi_m$$
    For the primary winding: $E_p = 4.44 f N_p \phi_m$
    For the secondary winding: $E_s = 4.44 f N_s \phi_m$

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21. **A single phase transformer is required to step down the voltage from 1100 V to 400 V at 50 Hz. The core has a cross-sectional area of 25 cm² and the maximum flux density is 5 Wb/m². Determine the number of turns of the primary and secondary windings.**
    Answer: Given, Primary voltage $V_p = 1100\ V$
    Secondary voltage $V_s = 400\ V$
    Frequency $f = 50\ Hz$
    Core cross-sectional area $A_c = 25\ cm^2 = 25 \times 10^{-4}\ m^2$
    Maximum flux density $B_m = 5\ Wb/m^2$
    $\therefore$ The maximum flux is given by: $\phi_m = B_m \times A_c$
    $\Rightarrow \phi_m = 5 \times 25 \times 10^{-4} = 0.0125\ Wb$
    The EMF equation of a transformer is:
    $$E = 4.44 f N\phi_m, \Rightarrow N = \frac{E}{4.44 f \phi_m}$$
    For the primary winding: $N_p = \frac{V_p}{4.44 f \phi_m}$
    $$\Rightarrow N_p = \frac{1100}{4.44 \times 50 \times 0.0125} \approx 396 \text{ turns (Ans)}$$
    For the secondary winding: $N_s = \frac{V_s}{4.44 f \phi_m}$
    $$\Rightarrow N_s = \frac{400}{4.44 \times 50 \times 0.0125} \approx 144 \text{ turns (Ans)}$$

22. **A 100 KVA single phase transformer has 300 turns in primary side and 30 turns in secondary side. The primary side is connected to a 11 kV, 50 Hz AC supply. Calculate:**
    i) secondary emf,
    ii) primary and secondary full load current and
    iii) maximum flux in the core.
    Solution: Given, Transformer rating 100 kVA
    Primary turns $N_1 = 300$
    Secondary turns $N_2 = 30$
    Primary voltage $V_1 = 11\ kV = 11000\ V$
    Frequency $f = 50\ Hz$
    i) Secondary EMF,
    Using turns ratio formula: $\frac{V_2}{V_1} = \frac{N_2}{N_1}$
    $\Rightarrow V_2 = V_1 \times \frac{N_2}{N_1} = 11000 \times \frac{30}{300} = 1100\ V$ (Ans)
    ii) Full load currents,
    $I = \frac{\text{Power}}{\text{Voltage}}$
    For primary current,
    $I_1 = \frac{100\ kVA}{11000\ V} = \frac{100000\ VA}{11000\ V} = 9.09\ A$ (Ans)
    For secondary current,
    $I_2 = \frac{100\ kVA}{1100\ V} = \frac{100000\ VA}{1100\ V} = 90.91\ A$ (Ans)
    iii) Maximum Magnetic Flux,
    $\phi_{max} = \frac{V_1}{4.44 f N_1} = \frac{11000}{4.44 \times 50 \times 300} = 0.165$ Wb (Ans)

23. **A 30 KVA single phase transformer has 500 primary turns and 30 secondary turns. The primary is connected to a 3300V, 50 Hz single phase supply. Calculate-**
    i) the maximum flux in the core;
    ii) secondary e.m.f;
    iii) primary and secondary full load current.
    Solution: Given, Transformer rating 30 kVA
    Primary turns $N_1 = 500$
    Secondary turns $N_2 = 30$
    Primary voltage $V_1 = 3300\ V$
    Frequency $f = 50\ Hz$
    i) Maximum Magnetic Flux,
    $\phi_{max} = \frac{V_1}{4.44 f N_1} = \frac{3300}{4.44 \times 50 \times 500} = 0.02973$ Wb (Ans)
    ii) Secondary EMF,
    Using turns ratio formula: $\frac{V_2}{V_1} = \frac{N_2}{N_1}$
    $\Rightarrow V_2 = V_1 \times \frac{N_2}{N_1} = 3300 \times \frac{30}{500} = 198\ V$ (Ans)
    iii) Full load currents,
    $$I = \frac{\text{Power}}{\text{Voltage}}$$
    For primary current,
    $I_1 = \frac{30\ kVA}{3300\ V} = \frac{30000\ VA}{3300\ V} = 9.09\ A$ (Ans)
    For secondary current,
    $I_2 = \frac{30\ kVA}{198\ V} = \frac{30000\ VA}{198\ V} = 151.52\ A$ (Ans)

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1. **When PN junction diode is in forward bias, by increasing the battery voltage-**
   (a) Circuit resistance increases
   (b) Current through P-N junction increases
   (c) Current through P-N junction decreases
   (d) None of the above happens.
   Explanation: When a PN junction diode is in forward bias, the positive terminal of the battery is connected to the p-type and the negative terminal to the n-type. This reduces the width of the depletion region, allowing more charge carriers to cross the junction.
   Answer: (b)

2. **A diode is a/an**
   (a) bidirectional device
   (b) unidirectional device
   (c) both (a) and (b)
   (d) none of the above
   Explanation: A diode is a unidirectional device because it allows current to flow in one direction only from the anode to the cathode when it is forward-biased. In reverse bias, the diode blocks the current.
   Answer: (b)

3. **Free electrons in p type material-**
   a) are majority carrier b) are minority carriers
   c) both (a) and (b) d) none of the above
   Explanation: In p-type material, the majority carriers are holes, and the free electrons are minority carriers. This is because p-type semiconductors are doped with acceptor atoms, creating more holes than electrons.
   Answer: (b)

4. **The number of valence electrons in the outermost shell of semiconductor are-**
   (a) 6 (b) 4 (c) 8 (d) 7
   Explanation: Semiconductors, such as silicon (Si) and germanium (Ge), belong to Group 14 of the periodic table. These elements have 4 valence electrons in their outermost shell, which allows them to form covalent bonds with neighboring atoms in a crystal lattice structure.
   Answer: (b)

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5. **The potential barrier of a silicon diode is**
   (a) 0.3 V (b) 0.7 V (c) 1 V (d) 5V.
   Explanation: The potential barrier of a silicon (Si) diode is typically 0.7V. This is the voltage required to overcome the built-in electric field of the p-n junction and allow current to flow in the forward direction.
   Answer: (b)

6. **At absolute temperature an intrinsic semiconductor has-**
   a) few free electrons
   b) few holes
   c) many holes and free electrons
   d) no holes and free electrons
   Explanation: At absolute zero temperature (0 K), an intrinsic semiconductor behaves like a perfect insulator all electrons are bound in the valence band and there is no thermal energy to excite them to the conduction band. Hence, no free electrons or holes are present.
   Answer: (d)

7. **FET is a ______ controlled device.**
   Explanation: A Field Effect Transistor (FET) is controlled by the voltage applied to its gate terminal, which controls the current flowing between the source and drain.
   Answer: voltage

8. **If the revers bias of gate in JFET is increased, the width of the conducting channel is _______.**
   Explanation: In a Junction Field Effect Transistor (JFET), increasing the reverse bias on the gate narrows the conducting channel, reducing the current flow.
   Answer: reduced

9. **The number of junctions in an NPN transistor are _________.**
   Explanation: An NPN transistor has two junctions: the base-emitter junction and the base-collector junction.
   Answer: 2

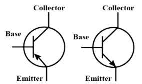
10. **The majority charge carrier in n-type material is _________.**
    Explanation: In n-type semiconductors, the majority charge carriers are free electrons, which are donated by donor impurities.
    Answer: electrons

11. Define the term Semiconductor.
Answer: A semiconductor is a material that has electrical conductivity between that of a conductor (like copper) and an insulator (like glass). Its conductivity can be controlled by temperature, impurities (doping), or an external electric field. Common semiconductors include silicon (Si) and germanium (Ge), which are widely used in electronic devices such as transistors, diodes, and integrated circuits.

12. Define the term Doping. Give examples of material for creation of p & n-type material.
Answer: Doping is the process of intentionally adding small amounts of impurities (called dopants) to a pure semiconductor (like silicon or germanium) to increase its conductivity by altering the number of charge carriers.
Types and Examples:
1) **n-type material** (adds electrons as majority carriers):
   - Formed by doping with pentavalent atoms (5 valence electrons)
   - Examples of dopants: Phosphorus (P), Arsenic (As), Antimony (Sb)
2) **p-type material** (adds holes as majority carriers):
   - Formed by doping with trivalent atoms (3 valence electrons)
   - Examples of dopants**:** Boron (B), Gallium (Ga), Indium (In)

13. Why is doping done on Semiconductor?
Answer: Doping is done on semiconductors to increase their conductivity by adding impurities. It creates n-type (electrons as majority carriers) or p-type (holes as majority carriers) semiconductors.

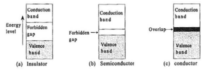
14. Draw the symbols of PNP and NPN transistor with proper notation.
Answer:



p-n-p n-p-n

PNP: The arrow on the emitter points inward (indicating conventional current flow).
NPN: The arrow on the emitter points outward (indicating electron flow direction).

15. **With respect to Energy band diagram differentiate between Conductors, Semiconductor & Insulator.**
Answer:
Insulators :
(i) In case of insulators, the energy gap between the conduction and valence bands is very large and the conduction band is practically empty.
(ii) When an electric field is applied to such kind of material, the conduction band remains to be empty. That is why no current flows through insulators.
Semiconductors:
(i) In case of semiconductor, the energy band structure is similar to insulator but in this case, the size of forbidden energy gap is quite smaller than that of the insulators.
(ii) When an electric field is applied to a semiconductor, the electrons in the valence band find it relatively easier to jump to the conduction band. So, the conductivity of semiconductors lies between the conductivity of conductors and insulators.
Conductors:
(i) In the case of a conductor there is no forbidden gap, and the valence and conduction energy bands overlap. For this reason, very large numbers of electrons are available for conduction.
(ii) Even when a small current is applied, conductors can conduct electricity.

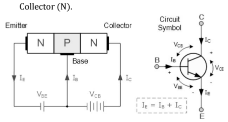


16. Why is transistor termed as current controlled device?
Answer: A transistor is termed a current-controlled device because the output current (collector current, I_C) is controlled by the input current (base current, I_B).
In a Bipolar Junction Transistor (BJT):
$$I_C = \beta . I_B$$
Where:
I_C = collector current (output)
I_B = base current (input)
β = current gain of the transistor
This means a small base current controls a much larger collector current, hence it's called current-controlled. This property makes BJTs useful for amplification and switching.

17. **Distinguish between an intrinsic and extrinsic semiconductor material.**
Answer:

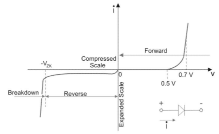
Intrinsic Semiconductor	Extrinsic Semiconductor
A pure semiconductor with no impurity atoms.	A semiconductor doped with impurity atoms to improve conductivity.
100% pure, only contains semiconductor atoms (e.g., Si, Ge).	Contains a small amount of impurity (dopant) added intentionally.
Equal number of electrons (n) and holes (p).	Carrier concentration is dominated by either electrons (n) or holes (p).
Low conductivity due to fewer charge carriers.	Higher conductivity due to increased free charge carriers from doping.
Fermi level lies near the center of the bandgap.	Fermi level shifts towards the conduction band in n-type and towards the valence band in p-type.
Conductivity increases with temperature.	Less affected by temperature due to dopant atoms.
Examples: Pure Silicon (Si), Pure Germanium (Ge).	Examples: Doped Silicon (Si) or Germanium (Ge) with P or As (n-type) or B or Al (p-type).

18. Explain with diagram the principle of operation of an NPN transistor.
Answer: An NPN transistor consists of two n-type semiconductors separated by a p-type layer, forming the structure: Emitter (N) – Base (P) – Collector (N).



Working Principle:
1) When the emitter-base junction is forward-biased, electrons (majority carriers in the N-type emitter) are injected into the base.
2) The base is very thin and lightly doped, so only a few electrons recombine with holes.
3) The majority of electrons pass through the base and reach the collector, which is reverse-biased and attracts them.
4) A small base current I_B controls a large collector current I_C.
5) This results in amplification, as the transistor transfers current from input to output.
$$I_E = I_C + I_B$$

19. Draw and explain the forward and reverse biased characteristics of P-N junction diode.
Answer:



Forward Bias Characteristics:
1) The depletion region narrows.
2) The built-in potential barrier is reduced.
3) Electrons from the N-side and holes from the P-side start moving toward the junction.
4) At a certain voltage (threshold voltage), a large current starts flowing.
5) Graph: The current exponentially increases after the threshold voltage is reached.

Reverse Bias Characteristics:
1) The depletion region widens.
2) The potential barrier increases.
3) Only a very small leakage current flows due to minority carriers.
4) At a high voltage (breakdown voltage), a large reverse current suddenly flows, leading to diode breakdown.
5) Zener Breakdown: Occurs in heavily doped diodes at low voltage.
6) Avalanche Breakdown: Occurs in lightly doped diodes at high voltage.
7) Graph: A very small reverse current flows until breakdown, after which the current increases sharply.

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1. **An ideal OPAMP is an ideal**
(a) current controlled current source
(b) current controlled voltage source
(c) voltage controlled voltage source
(d) voltage controlled current source
Explanation: An ideal operational amplifier (OPAMP) is a voltage-controlled voltage source (VCVS) because it amplifies the difference between its two input voltages (the inverting and non-inverting inputs) to produce a much larger output voltage.
Answer: (c)

2. **The expression of voltage gain in an inverting amplifier using ideal OP-APM is (R_f = feedback resistance, R_i = input resistance)-**
a) $-\frac{R_f}{R_i}$ b) $\frac{R_f}{R_i}$ c) $1+\frac{R_f}{R_i}$ d) $-\left(1+\frac{R_f}{R_i}\right)$
Explanation: For an ideal inverting amplifier using an op-amp, the voltage gain (A_v) is given by:
$$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$
The negative sign indicates a 180° phase shift between input and output.
Answer: (a)

3. **The expression of voltage gain in an non-inverting amplifier using ideal OP-APM is (R_f = feedback resistance, R_i = input resistance)-**
a) $-\frac{R_f}{R_i}$ b) $\frac{R_f}{R_i}$ c) $1+\frac{R_f}{R_i}$ d) $-\left(1+\frac{R_f}{R_i}\right)$
Explanation: In a non-inverting amplifier using an ideal op-amp, the voltage gain is given by:
$$A_v = \frac{V_{out}}{V_{in}} = 1+\frac{R_f}{R_i}$$
This configuration provides positive gain without phase inversion.
Answer: (c)

4. **The CMRR of an OP-Amp is the ratio of ______.**
Explanation: Common-Mode Rejection Ratio (CMRR) is the ratio of the differential gain to the common-mode gain, indicating how well the op-amp rejects common-mode signals.
Answer: Common Mode Gain to Differential Mode Gain

5. **An ideal OP-AMP has the following characteristics- (R_{in} = input resistance, A = open loop gain, R_{out} = output resistance)**
a) $R_{in} = \infty, A = \infty, R_{out} = 0$
b) $R_{in} = 0, A = \infty, R_{out} = 0$
c) $R_{in} = \infty, A = 0, R_{out} = \infty$
d) $R_{in} = 0, A = \infty, R_{out} = \infty$
Explanation: An ideal operational amplifier (OP-AMP) has the following characteristics:
- **Infinite input resistance ($R_{in} = \infty$):** This means no current flows into the input terminals of the OP-AMP.
- **Infinite open-loop gain ($A = \infty$):** The amplification of the input signal is infinite, meaning even a tiny difference between the inverting and non-inverting inputs would result in a huge output.
- **Zero output resistance ($R_{out} = 0$):** The OP-AMP can drive any load without any voltage drop across its output resistance.

Answer: (a)

6. **Name the two input terminals of OP-AMP.**
Answer: Two input terminals of an OP-AMP: Inverting (-) and Non-Inverting (+). These terminals receive input signals and determine the output behavior.

7. **What is a dependent Source?**
Answer: A dependent source is a voltage or current source whose value depends on another circuit variable.
Example: A current-controlled voltage source (CCVS) depends on an external current.

8. **What is virtual ground?**
Answer:
Virtual Ground: Virtual ground is a concept used in op-amp circuits, particularly in inverting configurations, where the voltage at the inverting terminal is approximately 0V (ground) even though it is not physically connected to the ground.
It helps in simplifying circuit analysis and ensures current flows correctly in feedback networks.

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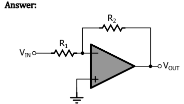
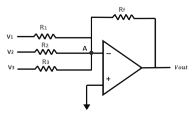
9. **Draw the circuit diagram of an inverting amplifier and derive the expression for output voltage.**
Answer:



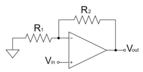
Derivation of Output Voltage Expression:
The non-inverting terminal (+) is grounded ($V_+ = 0$)
Due to virtual shorts concept, the inverting terminal (-) is also at 0 V ($V_- = 0$)
Apply Kirchhoff's Current law at node V_-.
Since the input impedance of an ideal Op-Amp is infinite, no current enters the Op-Amp. So, the current through R_1 must be equal to the current through R_2:
$$I_1 = I_2$$
Using Ohm's law: $\frac{V_{in}-0}{R_1} = \frac{0-V_{out}}{R_2}$
$\Rightarrow \frac{V_{in}}{R_1} = -\frac{V_{out}}{R_2}$
$$\Rightarrow V_{out} = -V_{in}\times\frac{R_2}{R_1}$$
The output is 180° out of phase with the input due to the negative sign.
The gain is given by $A_v = -\frac{R_2}{R_1}$

10. **Explain the op-amp acts as a adder circuit.**
Answer:



Working Principle of Op-Amp Adder:
The inverting terminal (-) is at virtual ground (due to the high gain of the Op-Amp). Since no current flows into the Op-Amp, all input currents sum up and pass through the feedback resistor R_f.
By Kirchhoff's Current Law (KCL) at the inverting terminal:
$$\frac{V_1}{R_1}+\frac{V_2}{R_2}+\frac{V_3}{R_3} = -\frac{V_{out}}{R_f}$$
For equal resistors ($R_1 = R_2 = R_3 = R$)
$$\Rightarrow V_{out} = -\frac{R_f}{R}(V_1+V_2+V_3)$$
The output is inverted (negative sign).

11. **Draw the circuit diagram of a non inverting amplifier with feedback using op-amp & derive the expression of gain.**
Answer:



Non-inverting amplifier circuit

V_{in} = input voltage, V_{out} = out voltage
R_1 = resistor connected to ground from inverting input
R_2 = feedback resistor
Since the op-amp has very high gain, the voltage at the inverting terminal (V_-) is equal to the voltage at the non-inverting terminal (V_+). $V_- = V_+ = V_{in}$
The current through R_1 is the same as the current through R_2 because of Kirchhoff's Current Law.
Applying Ohm's Law: The current through R_1 is:
$$I = \frac{V_-}{R_1} = \frac{V_{in}}{R_1}$$
The same current flows through R_2, so the voltage drop across R_2 is: $V_{out} - V_- = IR_2$
$\Rightarrow V_{out} - V_{in} = \frac{V_{in}}{R_1}.R_2$
$\Rightarrow V_{out} = \frac{V_{in}}{R_1}.R_2 + V_{in}$
$\Rightarrow V_{out} = V_{in}\left(1+\frac{R_2}{R_1}\right)$
$\therefore$ Voltage gain $A_v = \frac{V_{out}}{V_{in}} = 1+\frac{R_2}{R_1}$

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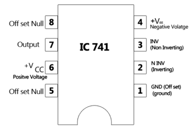
12. **Draw the pin diagram of IC 741 OPAMP.**
Answer:



Pin diagram of IC 741

Pin Number	Pin Name
1	Offset Null
2	Inverting Input (-)
3	Non-Inverting Input (+)
4	Vee (-) / GND
5	Offset Null
6	Output
7	Vcc (+)
8	NC (No Connection)

13. **Describe the function of each pin of IC 741 OPAMP**
Answer:
Pin Functions:

Pin No.	Function
1	Used for offset voltage adjustment (with Pin 5) to make output exactly zero when input is zero.
2	Input terminal where signal is applied with inverted phase (180° shift).
3	Input terminal where signal is applied with the same phase.
4	Connected to the negative power supply (e.g., –15V).
5	Used along with Pin 1 for zero offset voltage adjustment.
6	Provides the amplified output signal.
7	Connected to the positive power supply (e.g., +15V).
8	Not internally connected; usually unused.

14. **What are the characteristics of an ideal OP-AMP?**
Answer:
Characteristics of an Ideal Op-Amp:
An ideal operational amplifier (Op-Amp) is a theoretical device with perfect performance characteristics. Its key features include:
1) **Infinite Open-Loop Gain(A_v):** The voltage gain without feedback is infinite, meaning even a tiny input voltage difference produces a large output.
2) **Infinite Input Impedance (Z_{in}):** The input resistance is infinite, ensuring that no current flows into the input terminals.
3) **Zero Output Impedance (Z_{out}):** The output resistance is zero, allowing the Op-Amp to deliver maximum current to the load without voltage drop.
4) **Infinite Bandwidth:** The Op-Amp can amplify signals of any frequency without loss, meaning it has infinite gain at all frequencies.
5) **Infinite Common-Mode Rejection Ratio (CMRR):** An ideal Op-Amp completely rejects common-mode signal.
6) **Infinite Slew Rate:** The output can change instantaneously in response to input voltage changes, meaning there is no delay in response.
7) **Perfect Linearity:** The Op-Amp maintains a linear relationship between input and output across all operating conditions.

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1. **When the two input of a NAND gate is high, the output of the gate will be (a) high (b) low (c) toggle between high & low (d) none of the above**
Explanation: A NAND gate (NOT AND) gives a low output only when all its inputs are high. Its operation is the inverse of an AND gate.
Answer: (b)

2. **Which of these sets of logic gates are known as universal gates? (a) XOR, NAND, OR (b) OR, NOT, XOR (c) NOR, NAND, XNOR (d) NOR, NAND.**
Explanation: NAND and NOR gates are called universal gates because by combining multiple NAND or NOR gates, you can create all basic logic functions like AND, OR, and NOT.
Answer: (d)

3. **When two input of a NAND gate is high, the output of the gate will be (a) High (b) Low (c) Toggle between high & low (d) none of above.**
Explanation: A NAND gate is essentially an AND gate followed by a NOT gate. An AND gate outputs high only when all inputs are high. The NOT gate inverts the output of the AND gate, so a high output from the AND gate becomes a low output from the NAND gate, and vice versa.
Answer: (b)

4. **The number of flip flops required in a decade counter is- a) 2 (b) 3 (c) 4 (d) 10.**
Explanation: The number of flip-flops required is determined by: $2^n \geq 10$, where n is the number of flip-flops.
$2^3 = 8$ (not enough)
$2^4 = 16$ (sufficient)
Thus, 4 flip-flops are needed to count from 0 to 9 in a decade counter.
Answer: (c)

5. **The maximum possible number of states in a ripple counter with 5 flip-flops is- (a)32 (b) 15 (c) 10 (d) 5.**
Explanation: A ripple counter with n flip-flops can have a maximum of 2^n states.
For 5 flip-flops: $2^5 = 32$
Answer: (a)

6. **$\overline{A+B}$ is equivalent to- (a) A + B (b) A.B (c) $\bar{A}.\bar{B}$ (d) $\bar{A}+\bar{B}$**
Explanation: The given expression is $\overline{A+B}$. According to De Morgan's Law, the negation of the sum of two variables is equal to the product of their negations:
$$\overline{A+B} = \bar{A}.\bar{B}$$
Answer: (b)

7. **The output of AND gate will be HIGH only when both the input is ______.**
Explanation: In an AND gate, the output is HIGH (1) only when both inputs are HIGH (1). Otherwise, the output is LOW (0).
Answer: HIGH

8. **The full form of TTL is ________.**
Explanation: TTL is a type of logic family that uses transistors to perform logic operations, typically used in digital circuits.
Answer: Transistor-Transistor Logic

9. **Define the term Flip Flop?**
Answer: A Flip-Flop is a bistable multivibrator, which is a digital electronic circuit used to store one bit of data. It has two stable states (logic 0 and logic 1) and can change states based on input signals. Flip-flops are the fundamental building blocks of sequential logic circuits and are commonly used in memory elements, registers, and counters.

10. **Name any two types of Flip-Flops.**
Answer:
Two types of Flip-Flops: SR Flip-Flop and D Flip-Flop. Flip-flops store binary data and are used in memory and sequential circuits.

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11. State De-Morgan's Theorem.

Answer:

De Morgan's Laws in Boolean Algebra: De Morgan's Theorems provide an equivalence between AND, OR, and NOT operations. They state:

First Law: The complement of a product is equal to the sum of the complements.

$$\overline{A.B} = \overline{A} + \overline{B}$$

Proof:

A	B	A.B	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

Since $\overline{A.B}$ and $\overline{A} + \overline{B}$ produce the same output, 1st Law is proved.

Second Law: The complement of a sum is equal to the product of the complements.

$$\overline{A+B} = \overline{A}.\overline{B}$$

Proof:

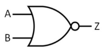
A	B	A + B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A}.\overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Since $\overline{A+B}$ and $\overline{A}.\overline{B}$ produce the same output, 2nd Law is proved.

12. Draw the NOR gate and write down its truth table.

Answer:

NOR gate:



Boolean Expression: $Z = \overline{A + B}$

Truth Table for NOR Gate:

Input A	Input B	Output $Z = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

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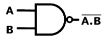
13. Draw the logic symbol and truth table of a 2-input NAND gate and EX-OR gate.

Answer:

2-input NAND gate:



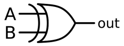
Logic Symbol: 2-input NAND gate

Boolean Expression: $Y = \overline{A.B}$

Truth Table:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

2-input EX-OR gate:



Logic Symbol: 2-input EX-OR gate

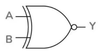
Boolean Expression: $Y = A.\overline{B} + \overline{A}.B = A \oplus B$

Truth Table of XOR Gate

A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

14. Draw the symbol of two input Ex-NOR gate.

Answer:



Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

2-input Ex-NOR gate

15. What are universal logic gates? Gives examples.

Answer:

Universal Gates: Universal gates are logic gates that can be used to implement any Boolean function without needing other types of gates.

Example: The NAND and NOR gates are called universal gates.

NAND and NOR Called Universal Gates because-

1. Functional Completeness: Any Boolean function can be designed using only NAND or NOR gates.

2. Replacement for Basic Gates: These gates can replicate the functions of AND, OR, and NOT gates.

3. Practicality in Circuit Design: Using only one type of gate (NAND or NOR) simplifies hardware design and reduces manufacturing costs.

16. Simplify the expression using Boolean algebra and realize the same by using logic gates.

$$Y = (A + \overline{B} + \overline{C}).(A + \overline{B}C)$$

Answer:

$Y = (A + \overline{B} + \overline{C}).(A + \overline{B}C)$

Using distribution:

$Y = A(A + \overline{B}C) + (\overline{B} + \overline{C})(A + \overline{B}C)$

Using Absorption Law $(A + A\overline{B}C = A)$:

$Y = A + (\overline{B} + \overline{C})(A + \overline{B}C)$

Expanding $(\overline{B} + \overline{C})(A + \overline{B}C)$:

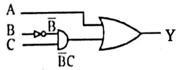
$Y = A + \overline{B}A + \overline{B}\overline{B}C + \overline{C}A + \overline{C}\overline{B}C$

Using Idempotent Law $(\overline{B}\overline{B} = \overline{B})$:

$Y = A + \overline{B}A + \overline{B}C + \overline{C}A + \overline{C}\overline{B}C$

Using Absorption $(A + \overline{B}A = A)$:

$Y = A + \overline{B}C$



17. What is the minimum number of NAND gates required to realize a XOR gate.

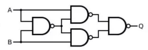
Answer: An XOR (Exclusive-OR) gate gives a high output (1) only when the inputs are different.

The Boolean expression for XOR is:

$$A \oplus B = A'B + AB'$$

This can be implemented using 4 NAND gates:

XOR



XOR gate by NAND gates

18. Give two applications of flip-flop.

Answer: Two applications of flip-flop are-

1) **Data Storage (1-bit Memory Cell):** Flip-flops are used to store a single bit of data (0 or 1). In digital systems, multiple flip-flops can be combined to create registers, which store multi-bit binary data in CPUs and memory systems.
2) **Counters and Timers:** Flip-flops are the building blocks of counters (e.g., ripple counters, synchronous counters) used in digital clocks, frequency dividers, and timers. They help in counting events or generating precise time delays.

19. What is decade counter?

Answer: A decade counter is a digital counter that counts from 0 to 9 (ten states) and then resets to 0 on the next clock pulse. It typically uses 4 flip-flops and is designed to output a binary sequence representing decimal digits. After reaching the binary equivalent of 9 (1001), it resets to 0000, making it useful for applications like digital clocks, frequency division, and event counting. Decade counters can be either synchronous or asynchronous depending on how the flip-flops are triggered.

{Prepared by NatiTute YouTube Channel}